

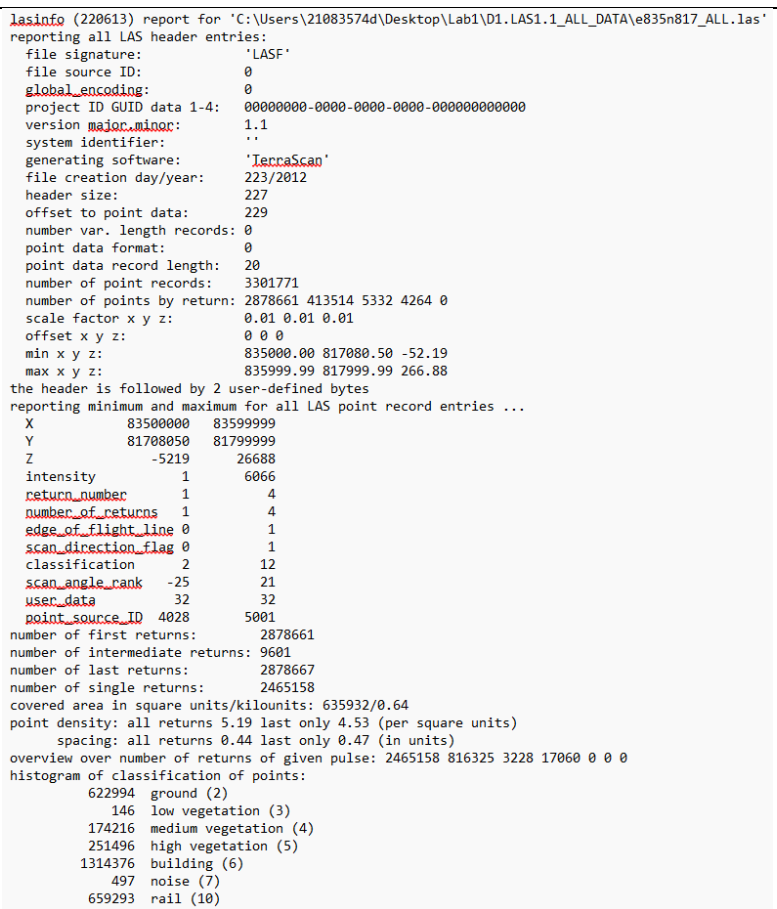
LSGI3262 Laser Scanning Lab 1

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1. Please retrieve the statistical information (e.g., spatial extent, of the above lidar dataset). [Refer to Demo1]

The following table shows the statistical information of e835n817_ALL.las. The file was retrieved from the CEDD Spatial Data Portal.

Key information	Screenshot from las header text file
- File signature: LASF - Generating Software: TerraScan - Number of points: 3301771 - Covered area: 635932 sq. units - File creation: day/year: 223th day of 2012 (10/8/2025) - Scale factor xyz: (0.01, 0.01, 0.01) - Offset xyz: (0, 0, 0)	 <pre> lasinfo (220613) report for 'C:\Users\21083574d\Desktop\Lab1\D1.LAS1.1_ALL_DATA\e835n817_ALL.las' reporting all LAS header entries: file signature: 'LASF' file source ID: 0 global_encoding: 0 project ID GUID data 1-4: 00000000-0000-0000-0000-000000000000 version major.minor: 1.1 system identifier: '' generating software: 'TerraScan' file creation day/year: 223/2012 header size: 227 offset to point data: 229 number var. length records: 0 point data format: 0 point data record length: 20 number of point records: 3301771 number of points by return: 2878661 413514 5332 4264 0 scale factor x y z: 0.01 0.01 0.01 offset x y z: 0 0 0 min x y z: 835000.00 817080.50 -52.19 max x y z: 835999.99 817999.99 266.88 the header is followed by 2 user-defined bytes reporting minimum and maximum for all LAS point record entries ... X 83500000 83599999 Y 81708050 81799999 Z -5219 26688 intensity 1 6066 return_number 1 4 number_of_returns 1 4 edge_of_flight_line 0 1 scan_direction_flag 0 1 classification 2 12 scan_angle_rank -25 21 user_data 32 32 point_source_ID 4028 5001 number of first returns: 2878661 number of intermediate returns: 9601 number of last returns: 2878667 number of single returns: 2465158 covered area in square units/kilounits: 635932/0.64 point density: all returns 5.19 last only 4.53 (per square units) spacing: all returns 0.44 last only 0.47 (in units) overview over number of returns of given pulse: 2465158 816325 3228 17060 0 0 0 histogram of classification of points: 622994 ground (2) 146 low vegetation (3) 174216 medium vegetation (4) 251496 high vegetation (5) 1314376 building (6) 497 noise (7) 659293 rail (10) </pre>

Max XYZ	835999.99	817999.99	266.88
Min XYZ	835000.00	817080.50	-52.19
XYZ spatial extent (unit)	999.99	919.49	319.07

2. Please retrieve the mean point spacing and density of an airborne lidar point cloud. If you have to rasterize this airborne lidar data with a specific spatial resolution, which value you are going to set. [Refer to Demo1]

From e835n817_ALL.las header, we have:

Mean point spacing = 0.44 units

Mean point density = 5.19 per square units

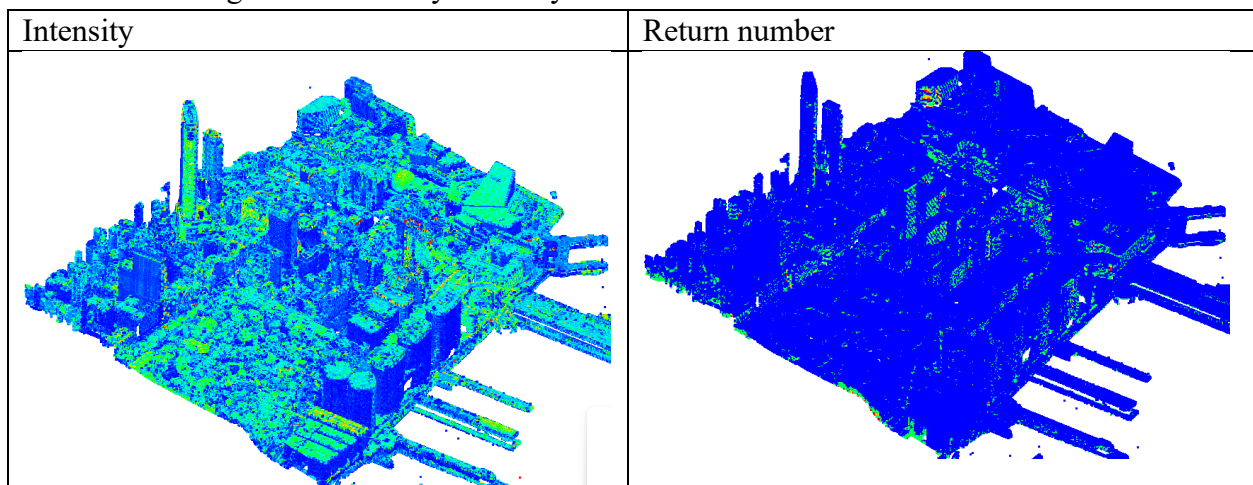
When rasterizing this airborne LiDAR data, the spatial resolution should be set at least mean point spacing. This makes each raster contains one point on average, and avoid empty data. For example, since the mean point spacing is 0.44 units, it would be appropriate to set the spatial resolution to 0.5 units. It maximizes the number of raster cells while being a unit fraction ($\frac{1}{2}$) makes it easier for calculations. If a coarser raster were acceptable, spatial resolution of 1 unit could be used, further ensuring the cells are non null values while being an integer.

3. How many different number of returns in the las file? Which land cover object(s) come with multiple returns and only single return? Please explain the reason behind. [Refer to Demo2]

From the .las file, there are 4 different number of returns (1, 2, 3, or 4 points).

number of points by return: 2878661 413514 5332 4264 0

To visualize the correlation between land cover objects and return number, the following figures show the scanning data colored by intensity and return number.



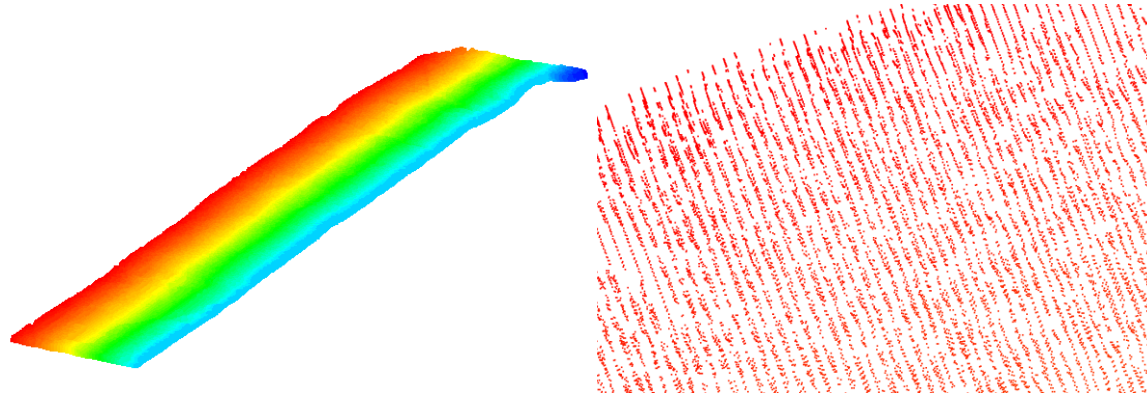
It appears that the most areas come with single return, while vegetation and the side of the buildings are more likely to have multiple returns. For most areas, the light is likely unobstructed and does not get partially reflected. Therefore, single returns are associated with flat, unobstructed surfaces.

For vegetation, this is likely due to the footprint size of the laser being larger than the trees, the light could be partially reflected from the leaves and branches before reaching the terrain.

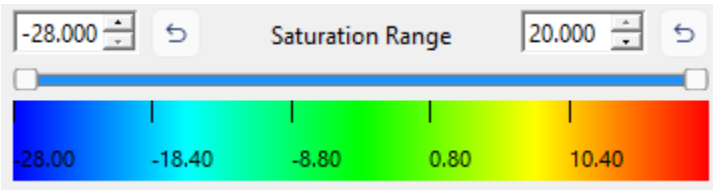
Similarly, for the side of the buildings likely have glass windows installed, and the footprint size of the laser could be larger than the windows, and glass is a semi-transparent surface, the light could be partially reflected from the buildings, then reflected from the interior of the building. Therefore, multiple returns are associated with complex, partially reflecting surfaces.

4. Please illustrate the scanning pattern of different lidar datasets and describe them based on the use of scan angle. [Refer to Demo3]

Figure of Airbone_LiDAR_1064.las scanning angle rank distribution (overview and closeup)

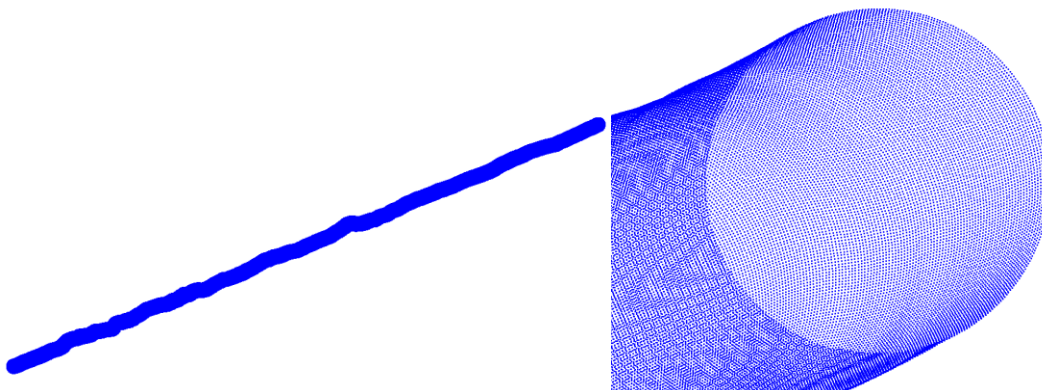


Legend of Airbone_LiDAR_1064.las scanning angle



From the figures, it appears that the magnitude of the scanning angle is the smallest at the middle (0) and largest at the sides (as low as -28 and as high as 20). The scans appears to be parallel lines. Therefore this scan likely uses oscillating mirror and scans side to side.

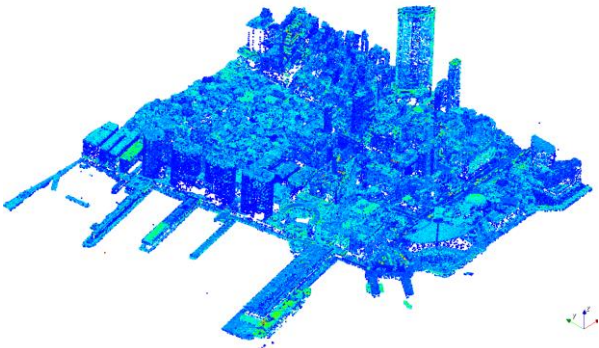
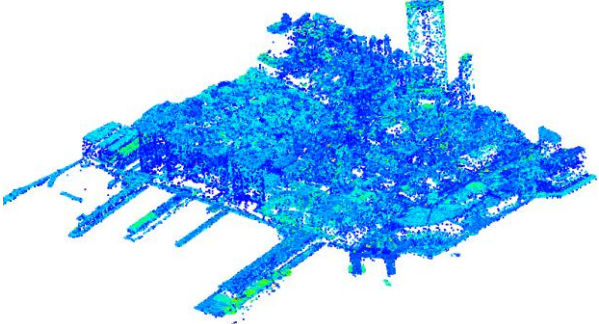
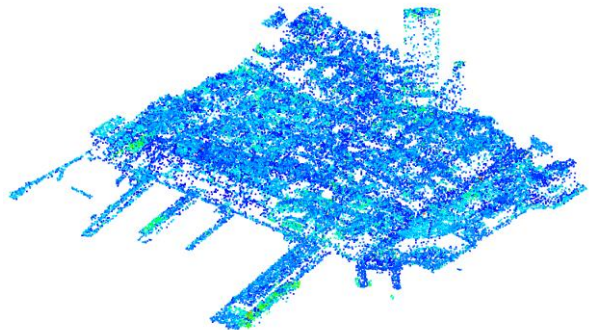
Figure of Bathymetric_liDAR_IR.las scanning angle rank distribution (overview and closeup)

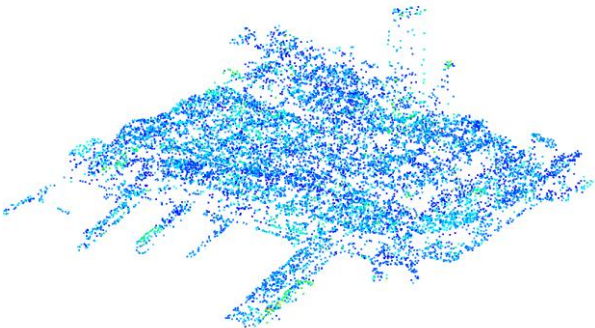


From the figures, it appears that the scanning angle remains constant throughout the scan, and the scanning pattern appears form circular shapes. Therefore this scan likely uses rotating wedge prism and scans in a circular motion.

5.Please perform down sampling of the dataset with different ratios (e.g., 4 different scales) and provide the print screen. Explain the loss of details, if found, in any data objects. [Refer to Demo4]

Total points: 3301771

Downsampled image	Description
	Downsampling factor: 4 Total points: 825443 The buildings and street boundaries, including the windows are clearly visible.
	Downsampling factor: 16 Total points: 206361 The windows blend with the buildings, the buildings and street boundaries fairly visible.
	Downsampling factor: 64 Total points: 51590 Only a few major buildings visible, streets and smaller building details are lost.

	Downsampling factor: 256 Total points: 12897 Most details are lost. Buildings and street boundaries blend together. The coastline still visible.
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